# REPORT — Bridge C order3c (analytic O(1) term)

PREREG sha256 cd14c9ab…d8ad (unchanged). Scripts are hashed in scripts/SHA256SUMS.

## Statement (proved below)

Notation is as in order2 entry 107. For a piece with basis U, offset c, companion (α,β), x = h/m and N = 2m: - $\tilde{Q}(s) = p\,s1^2 + q\,s1s2 + r\,s2^2$, positive on the closed quadrant, with $q > 0$ and $\Delta = q^2 - 4pr > 0$; - $E(s) = \tilde{Q}(s) + l1s1 + l2s2 + Q0$, with $Q0 = Q(\alpha+c)$; - f is N-periodic on $Z^2_{\geq 0}$, with $|f| = 1$.

Then, as $t \to 0+$:

$$\Sigma_{s\in Z^2_{\geq 0}} f(s)e^{-tE(s)} = t^{-1}\mathrm{mean}(f)Ar + t^{-1/2}A1_P + A2_P + O(t^{1/2}),$$

$$A2_P = \mathrm{mean}(f)(\tfrac{1}{2}K(l) - Q0Ar) + W_1\, l2/(2r) + W_2\, l1/(2p) + (q/4)(V_1/r + V_2/p) + C,$$

where: - $K(l) = \iint_{R^2\geq 0}(l1u1 + l2u2)^2 e^{-\tilde{Q}(u)}du$; - $W_i = N^{-1}\Sigma_z f(z)B1(z_i/N)$; - $V_i = \Sigma_z f(z)B2(z_i/N)$; - $C = \Sigma_z f(z)B1(z1/N)B1(z2/N)$; - z runs over $\{0,\dots,N-1\}^2$.

For a wall $\{c + bu : b \geq 0\}$, with $l_u = B(u, \alpha+c)$, the constant term is $-\mathrm{mean}(f)\cdot l_u/(2Q(u)) - \Sigma_{b<N} f(b)B1(b/N)$.

Hence A2 = Σ_pieces σ_P A2_P + Σ_walls σ_w(wall constant) for every companion and every x. This uses the same signed piece/wall decomposition as entry 107.

In polar coordinates, $K_{ij} = \tfrac{1}{2}\int_0^{\pi/2} e_i(\theta)e_j(\theta)/\varphi(\theta)^2\, d\theta$, with $e = (\cos\theta, \sin\theta)$ and $\varphi(\theta) = \tilde{Q}(e(\theta))$. ($Ar = \tfrac{1}{2}\int d\theta/\varphi$ was checked against the entry-107 closed form to 1e-12.)

## Proof

**(i) One-variable Euler–Maclaurin.** Let h be $C^3$ on $[0,\infty)$ with $h, h', h'', h'''$ integrable, and $y0 \in [0,1)$. Then

$$\Sigma_{m\geq 0} h(N(y0+m)) = N^{-1}\int_0^\infty h - B1(y0)h(0) - (N/2)B2(y0)h'(0) + R_3,$$

with $|R_3| \leq (N^2/6)\cdot\sup|\tilde{B}3|\cdot\int_0^\infty|h'''|$. This is the standard periodic-Bernoulli form of Euler–Maclaurin applied to $H(y) = h(Ny)$ on $[y0,\infty)$, written relative to the base point 0.

**(ii) Scaling.** Put $F_t(s) = e^{-tE(s)}$. On the closed quadrant $\tilde{Q}(s) \geq \kappa|s|^2$ for some $\kappa > 0$, since $q > 0$ and $p, r > 0$ for all pieces. So: - every $\partial^a F_t$ is bounded by $C_a\, t^{|a|/2}(1 + (\sqrt{t}|s|)^{|a|})e^{-t\kappa|s|^2/2}$; - hence $\int_0^\infty|\partial_i^k F_t(s)|\, ds_i = O(t^{(k-1)/2})$ uniformly in the other variable; - $\iint|\partial^a F_t| = O(t^{|a|/2 - 1})$.

**(iii) Iteration.** Fix a residue class $z = (\alpha',\beta')$. Apply (i) in s2 (step N, offset $\beta'/N$), then apply (i) in s1 to each resulting term.

Main terms that survive at order $t^0$ or lower: - $N^{-2}\iint F_t$; - $-N^{-1}B1(\beta'/N)\int_0^\infty F_t(s,0)ds$ and $-N^{-1}B1(\alpha'/N)\int_0^\infty F_t(0,s)ds$; - $-\tfrac{1}{2}B2(\beta'/N)\int_0^\infty\partial_2 F_t(s,0)ds$ and $-\tfrac{1}{2}B2(\alpha'/N)\int_0^\infty\partial_1 F_t(0,s)ds$; - $+B1(\alpha'/N)B1(\beta'/N)F_t(0,0)$.

Cross terms and remainders: - $B1\cdot(N/2)B2\cdot\partial F_t(0,0) = O(t^{1/2})$; - $(N/2)B2(\alpha'/N)\cdot\partial_1 F_t(0,0) = O(t^{1/2})$; - the $R_3$ terms are $O(N^2\int|\partial^3 F_t|) = O(t^{1/2})$, by (ii); - the s-integrals of the s2-remainders are $O(N^2\cdot\iint|\partial_2^3 F_t|\cdot\dots) = O(t^{1/2})$, by (ii).

So every neglected term is $O(t^{1/2})$.

**(iv) Each surviving term to $O(t^{1/2})$.** - *Area term.* Substitute $u = \sqrt{t}\,s$: $\iint F_t = t^{-1}\iint e^{-\tilde{Q}(u) - \sqrt{t}L(u) - tQ0}$. Taylor's theorem with the majorant from (ii) gives $t^{-1}Ar - t^{-1/2}(l1J_a + l2J_b) + (\tfrac{1}{2}K(l) - Q0Ar) + O(t^{1/2})$. - *Edge integrals.* $\int_0^\infty F_t(s,0)ds = \tfrac{1}{2}\sqrt{(\pi/(tp))} - l1/(2p) + O(t^{1/2})$, since $\int_0^\infty u e^{-pu^2} = 1/(2p)$. Likewise $\int_0^\infty F_t(0,s)ds = \tfrac{1}{2}\sqrt{(\pi/(tr))} - l2/(2r) + O(t^{1/2})$. - *Derivative integrals.* $\int_0^\infty\partial_2 F_t(s,0)ds = -t\int(qs + l2)e^{-t(ps^2 + l1s + Q0)}ds = -q/(2p) + O(t^{1/2})$. Similarly $\int_0^\infty\partial_1 F_t(0,s)ds = -q/(2r) + O(t^{1/2})$. - *Corner.* $F_t(0,0) = 1 + O(t)$.

**(v) Summing.** Multiply by f(z) and sum over z. The class factors give: - mean(f) from the $N^{-2}$ term; - $W_i$ from the $N^{-1}B1$ terms; - $V_i$ from the B2 terms; - C from the corner term.

Collecting the t^0 parts gives A2_P. The t^{−1} and t^{−1/2} parts reproduce entry 107, which is a consistency check.

**Walls.** These are the one-variable case of (i) with h(b) = e^{−tQ(α+c+bu)}, and the constant term comes from the same edge-integral expansion.

**Assembly.** Summing over the signed pieces and walls gives A2. The decomposition is the finite signed identity of entry 107 (orthant, 2×21 pieces, 4 half-weight walls), so the O(t^{1/2}) remainders add up to O(t^{1/2}). ∎

**Category:** A. This is a finite formula with an explicit remainder order. The constants in O(t^{1/2}) depend on N and the piece and are not made numerically explicit.

## Sign control (gating)

The formula was recomputed with the corner term C sign-flipped. Of the 9 x values, 7 have |C_total| > 0.05; x = 1/2 has C_total = 0. At every x with |C_total| > 0.05, the flipped formula fails: - err_flipped from 0.25 to 2.92, all > 1e-2; - the check discriminates. **Passed.**

## Map N (Category B): brute-force series S_{a,0}

Grid: t_j = 2^{−j}·0.5/(297.5(2m)²), j = 0..7. Fit on {1, t^{1/2}, t, t^{3/2}}.

| x | A2 (formula) | c0 (fit) | rel. err | verdict |
|---|---|---|---|---|
| 1 | −0.3232819 − 0.4449593i | −0.3232819 − 0.4449593i | 4.4e-10 | pass |
| 1/2 | ≈0 (1.6e-15) | 2.1e-8 | 2.1e-8 | pass |
| 1/3 | 0.2436643 + 0.3827023i | 0.2436643 + 0.3827023i | 2.1e-9 | pass |
| 2/3 | −0.4528349 + 0.0278183i | same to 1e-9 | 1.7e-9 | pass |
| 1/4 | 2.4890282 + 0.2339631i | 2.4890281 + 0.2339631i | 5.1e-8 | pass |
| 1/5 | 1.1047360 − 0.1395606i | same to 1e-8 | 5.0e-9 | pass |
| 1/6 | 2.3314379 + 1.8879612i | 2.3314376 + 1.8879609i | 1.4e-7 | pass |
| 1/7 | 1.0780550 − 0.3522517i | same to 1e-8 | 8.5e-9 | pass |
| 3/7 | 0.5314255 + 0.3597670i | same to 2e-8 | 1.7e-8 | pass |

**Verdict: pass, 9/9.** The largest relative error is 1.4e-7 (threshold 1e-3). These A2 values also reproduce the descriptive order3 Map 0 estimates.

# order3c notes (descriptive)

- **Ingredients of A2.** A2 has four kinds of term: a 2-D moment K(l) of the drift, edge-linear B1 terms, B2 terms weighted by the cross coefficient q, and a corner term B1·B1. The corner and B2 terms are elementary (Dedekind-sum-type) finite sums. Only K(l) and Ar involve transcendental integrals.
- **Closure not tested.** The S-closure of A2 was not tested. It is the natural next preregistration: does A2 (or A2 minus a completion term) satisfy a weight-0 vector law?
- **Grid.** The grid was scaled to the denominator and to Q(21,−1) = 595/2, and no post-hoc adjustment was needed.
- **Process.** Cutoff X = 36/t_min (truncation < e^{−36}). Runtime per x was up to ~30 min, single-threaded, under nice.